Embedded C and CMSIS

Introduction to ARM Cortex-M Microprocessors — Chapter 4

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Why This Chapter Matters

This chapter joins the two halves of the book. You know C. You know how the processor and memory are organized. Now we connect them: a C variable becomes a register, a C structure becomes a peripheral, and one assignment becomes a voltage on a pin. This is where hardware first moves.

After this chapter, you can:

1. Predict the assembly the compiler generates for simple C.
2. Explain the CMSIS components and which one does register-level work.
3. Show how a C structure overlays a peripheral's registers.
4. Justify `volatile` with two optimizations it prevents.
5. Apply the set, clear, toggle and test idioms.
6. Write a bare-metal CMSIS program that drives a GPIO pin.

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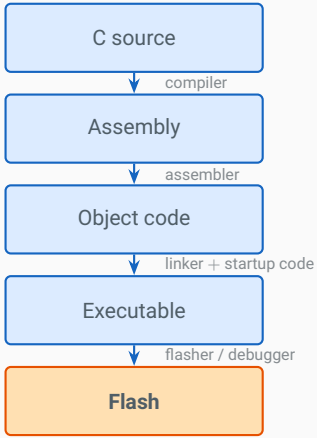
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What the Compiler Does



Each step is a separate program. You can inspect each one. This is how you find out what your C code actually became.

Why you learned assembly first

You will write C for the rest of this book. But when something goes wrong, the debugger shows you instructions, not C.

Chapters 5–7 exist so this disassembly window is readable, not frightening.

The linker adds startup code. It sets the stack pointer, zeroes your global variables, and calls `main()`, all before any line you wrote runs.

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C or Assembly?

	C	Assembly
Development speed	Fast	Slow
Readability	Good	Poor
Portability	Across ARM families	None
Code size	Good, with a modern compiler	Smallest possible
Hardware access	Through pointers and CMSIS	Direct
Typical use	Almost everything	Startup, ISR entry, hand-tuned DSP

What C gives you, and what it does not

C gives you the same hardware access as assembly, with far less bookkeeping. What it does *not* give you is protection. There is no runtime check to catch a bad pointer.

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What CMSIS Is

Cortex **M**icrocontroller **S**oftware **I**nterface **S**tandard. This is ARM's layer between your C code and the silicon. It works the same way across different chip vendors.

What it gives you

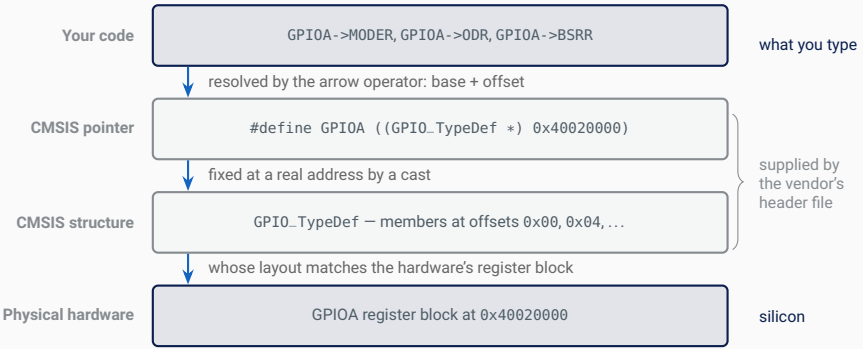
- A struct per peripheral, laid out on the real register offsets
- A pointer to each peripheral at its base address
- Named constants for bit fields
- Core functions: `__enable_irq()`, `NVIC_EnableIRQ()`, `__WFI()`

You need just one header: `#include "stm32f446xx.h"`

Why it matters

- The same names across every vendor's Cortex-M
- Readable: `GPIOA->ODR` rather than `*(volatile uint32_t*)0x40020014`
- The struct layout does the address arithmetic for you

The CMSIS Chain Silicon at the bottom, your code at the top



You only need to think about the top and bottom layers: the hardware exists, and your code names its registers. The two middle layers are a struct and a cast. They are written once, in a header file that you never edit.

A Peripheral as a Struct

```
typedef struct {
  __IO uint32_t MODER;
  __IO uint32_t OTYPER;
  __IO uint32_t OSPEEDR;
  __IO uint32_t PUPDR;
  __IO uint32_t IDR;
  __IO uint32_t ODR;
  __IO uint32_t BSRR;
  __IO uint32_t LCKR;
  __IO uint32_t AFR[2];
} GPIO_TypeDef;
```

Each member is one 32-bit register, in hardware order. A `uint32_t` is 4 bytes, so member *n* sits at offset 4*n*. These are exactly the offsets given in the reference manual.

`__IO` expands to `volatile`. CMSIS marks every register for you.

`AFR[2]` uses Chapter 3's array indexing: two real registers, reached as `AFR[0]` and `AFR[1]`.

From Name to Address

```
#define GPIOA_BASE  0x40020000UL

#define GPIOA \
  ((GPIO_TypeDef *) GPIOA_BASE)
```

A cast turns a plain number into a pointer to the struct. After this, the compiler computes every register's address for you.

`ODR` is the sixth member, so its offset is $5 \times 4 = 20 = 0x14$. This gives `0x40020014`, the same address as in Chapter 1.

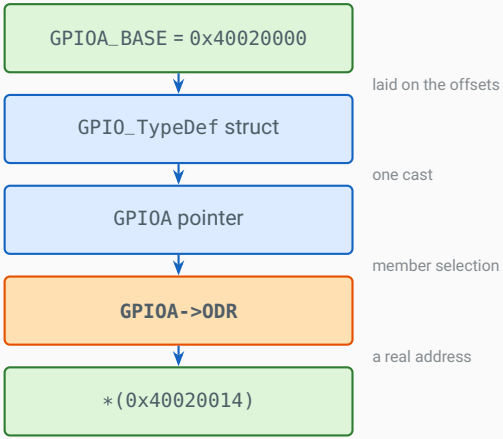


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1. Caching a value in a register

```
while ((*status & 1) == 0)
    ;   // waits forever without
        // volatile
```

Without `volatile`, the compiler reads the address once, and never checks it again.

2. Discarding a “redundant” write

```
*reg = 1;
*reg = 0;   // without volatile the
            // first write is deleted
```

Both writes matter if the address is a pulse output.

The rule

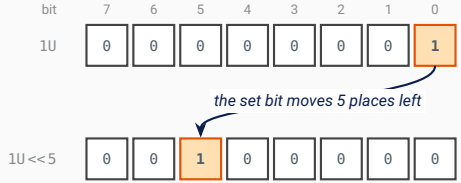
Every pointer to a hardware register must be `volatile`. So must every variable shared with an interrupt handler.

A compiler may assume that memory only changes when the program changes it. For a status register updated by hardware, that assumption is wrong.

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Building a Mask: The Shift

Every bit idiom starts with `1U << n`. Begin with `1U`: only bit 0 is set. Shift it left by *n*, and the only set bit is now bit *n*. This value is called a **mask**.



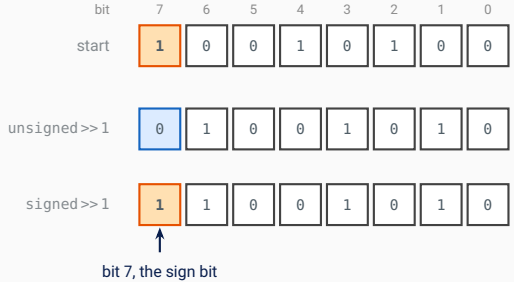
zeros fill in behind it, so exactly one bit of the mask is set.

The shift count *is* the bit number

`1U << 5` selects bit 5, not bit 6. Writing the pin number directly in the shift keeps the code readable, and easy to change for a different pin.

Why the U Suffix

A plain `1` is signed. Its top bit is the sign bit. C treats signed and unsigned shifts differently, and this difference matters exactly here.

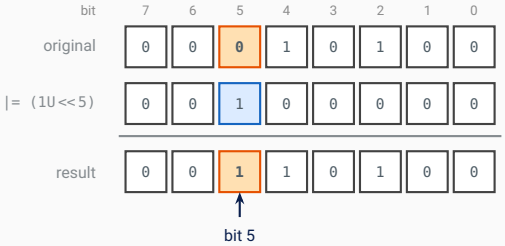


*a **0** is shifted in: the value simply halves.*

*the **sign bit is copied**: a negative value stays negative.*

`1 << 31` is undefined behaviour on a 32-bit `int`. `1U << 31` is fully defined, and gives `0x80000000`. A register is 32 switches, not a signed number. Always write `U`.

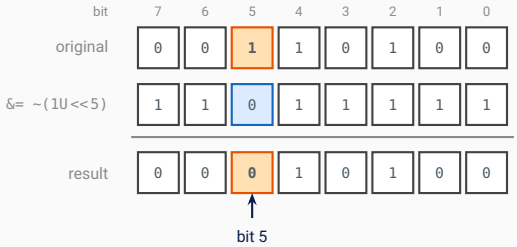
Setting a Bit: |=



OR with 1 forces a bit to 1;
OR with 0 leaves it alone.

`GPIOA->ODR |= (1U << 5);` — only the masked bit changes; every other bit is OR-ed with 0 and passes through untouched.

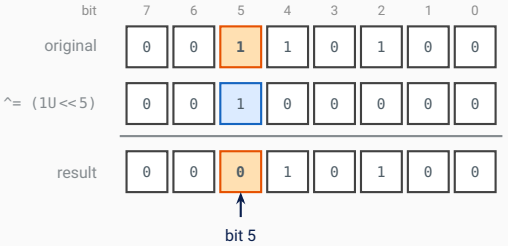
Clearing a Bit: &= with an Inverted Mask



AND with 0 forces a bit to 0;
AND with 1 leaves it alone.

`GPIOA->ODR &= ~(1U << 5);` — the tilde makes every bit except bit 5 a 1, so only bit 5 is forced low. The compiler emits `BIC`.

Toggling a Bit: ^=



XOR with 1 inverts a bit;
XOR with 0 preserves it.

`GPIOA->ODR ^= (1U << 5);` — the same statement run twice returns the register to its starting value, which is why one line suffices to blink an LED.

Multi-Bit Fields: Clear, Then Set

OR alone cannot clear a bit. Say `MODER` bits 11:10 read `11`, and you OR in `01`. The result is still `11`. You must clear the field first, before setting it.



OR alone cannot clear a bit, so a multi-bit field must be zeroed first…

…and only then written with the value you want.

The Four Bit Operations

Goal	In C	Compiles to
Set bit *n*	`REG |= (1U << n);`	`ORR`
Clear bit *n*	`REG &= ~(1U << n);`	`BIC`
Toggle bit *n*	`REG ^= (1U << n);`	`EOR`
Test bit *n*	`if (REG & (1U << n))`	`TST`

Read–modify–write

A peripheral register holds settings for many pins. You must keep the bits you are not changing. `|=` and `&=` do this automatically. A plain `=` would wipe out every other setting.

A multi-bit field needs a mask: first clear it with `&= ~(3U << pos)`, then set it with `|= (value << pos)`. The mask is always "all ones for this field."

A Complete Bare-Metal Program

```c
#include "stm32f446xx.h"

int main(void)
{
  // 1. Enable the clock for GPIOA
  RCC->AHB1ENR |= (1U << 0);

  // 2. Set PA5 as an output
  GPIOA->MODER &= ~(3U << (5*2));
  GPIOA->MODER |=  (1U << (5*2));

  // 3. Toggle PA5 forever
  while (1) {
    GPIOA->ODR ^= (1U << 5);
    for (volatile int i = 0;
         i < 100000; i++);
  }
}
```

Every idea in this chapter

- Struct-based register access
- Bit masking and shifting
- Read–modify–write
- Clock enable, then configure

Why `5*2`? `MODER` gives each pin two bits, so pin 5's bits start at position 10.

Why `volatile` on `i`? Without it, the optimiser deletes the whole delay loop.

The Clock Must Come First

```
RCC->AHB1ENR |= (1U << 0);
```

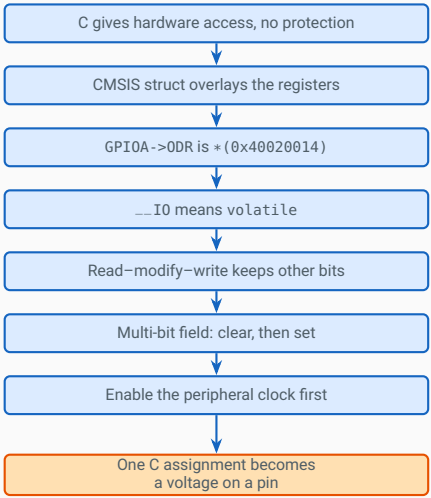
- Every peripheral starts with its clock disabled, to save power.
- A peripheral with no clock does not respond to writes. They are simply lost.
- `RCC` is the Reset and Clock Control peripheral. GPIOA is bit 0 of `AHB1ENR`.

The classic first bug

You write the GPIO configuration. The code compiles and runs. But nothing happens. No fault, no error: the writes went nowhere, because the clock was off.

Enable the clock first. Always.

Summary



Terminology covered

- CMSIS, device header
- Struct overlay
- `volatile`, `__IO`
- Set, clear, toggle, test
- Peripheral clock enable

Next: Chapter 5

ARM Assembly Fundamentals

We read what this C compiles to.